MUHAMMAD ASAD FEROZI

Applicant, MS in Computer Science (Artificial Intelligence) · Fall 2027

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RESEARCH INTERESTS

Applied machine learning and large language models; retrieval-augmented generation and grounded question answering; trustworthy and secure AI; computer vision for assistive technology; real-time and distributed systems for AI applications.

I hold a BS in Information Technology (CGPA 3.78/4.00) and have two years of industry experience building production LLM features for enterprise search and summarisation. I am seeking a research-based master's to study these systems rigorously.

EDUCATION

Bachelor of Science in Information Technology (4 years, 134 credit hours) 2021 – 2025

Punjab University College of Information Technology (PUCIT), University of the Punjab, Lahore **CGPA 3.78 / 4.00 · 85.40%**

- Final-semester GPA 3.94; first division. English was the medium of instruction.
- Capstone project: *Parental Eye: Child Tracking System*. Grades: Capstone I A (85), Capstone II A (90).

Design & Analysis of Algorithms	A · 90	Artificial Intelligence	A · 85	Computer Organization & Design	A · 93
Discrete Structures	A · 91	Linear Algebra	A · 85	Computer Networks	A · 90
Data Structures & Algorithms	A · 85	Calculus & Analytical Geometry	A · 86	Cyber Security	A · 85
Programming Fundamentals	A · 94	Probability & Statistics	A− · 80	Information Security	A− · 82
Software Quality Assurance	A · 89	Machine Learning	B · 73	Operating Systems	A− · 81

Intermediate (FSc Pre-Engineering), Board of Intermediate & Secondary Education, Multan 2021

Punjab College for Boys, Burewala **1071 / 1100 (97.4%) · A+**

Matriculation (Science), Board of Intermediate & Secondary Education, Multan 2019

Govt. Higher Secondary School Sheikh Fazal, Burewala **1049 / 1100 (95.4%) · A+**

RESEARCH AND ACADEMIC PROJECTS

Parental Eye: Real-Time Child Tracking System · Final-year capstone, PUCIT Oct 2024 – Jun 2025

React · Node.js · PostgreSQL · WebSockets · MQTT

- Built a full-stack real-time monitoring system with geofencing, location history, push notifications and safe-zone alerts.
- Kept devices and the dashboard in sync in real time using WebSockets and MQTT.

AI Lawyer: Retrieval-Augmented Legal Assistant Jun 2024 – Oct 2024

React · Node.js · OpenAI models · Pinecone vector database

- Built a chatbot that answers questions on Pakistan's legal framework, grounding language-model answers in semantic search over legal texts.

Be My Eye: Obstacle Detection for the Visually Impaired Jan 2024 – Apr 2024

Python · Computer vision · Roboflow · Google Colab

- Trained a real-time obstacle-detection model reaching 90% accuracy, 92% precision and 91% recall.

PROFESSIONAL EXPERIENCE

Full Stack Engineer, INTECH Automation and Intelligence, Lahore Jun 2025 – Present

- Build LLM features for **EmpowerGPT**, an enterprise platform for search, document summarisation and workflow automation over structured and unstructured data.
- Develop full-stack applications in React, Flask and PostgreSQL across 3+ enterprise projects.
- Built RESTful APIs serving 10,000+ requests a day; PostgreSQL indexing and caching cut average response time by 35%.
- Containerised services with Docker behind Nginx, reaching 99.5% uptime and cutting deployment time by 40%.

Full Stack Developer, Webox Systems, Lahore — Jul 2024 – Jun 2025

- Developed **Euthyna**, a sustainability analytics platform that automates ESG gap analysis and compliance checks (PostgreSQL, Express, React, Node.js).
- Code-splitting, database indexing and API optimisation cut page load times by 30%; reusable, accessible React components raised user engagement by 25%.
- Designed PostgreSQL schemas across 15+ tables, improving query efficiency by 30%.

Full Stack Development Fellow, Webox Systems, Lahore — Feb 2024 – May 2024

- Three-month fellowship building web applications with React, Node.js, Express and PostgreSQL.

Frontend Development Fellow, Bytewise Limited (remote) — Mar 2023 – Jul 2023

- Three-month fellowship; built 5+ projects with HTML, CSS, JavaScript, Tailwind CSS and React.

TECHNICAL SKILLS

Languages	Python, JavaScript (ES6+), TypeScript, SQL, C, C++, Java
AI and Machine Learning	Large language models (OpenAI API), retrieval-augmented generation, vector databases (Pinecone), NLP, computer vision (Roboflow)
Systems and Back End	Node.js, Express, Flask, FastAPI, REST, GraphQL, WebSockets, MQTT, microservices, caching, load balancing
Data	PostgreSQL, MySQL, MongoDB, Redis; schema design, indexing and query optimisation
Front End	React, Next.js, Redux, HTML5, CSS3, Tailwind CSS, accessibility
DevOps and Cloud	Docker, Nginx, Git, GitHub, CI/CD, Microsoft Azure, Linux

CERTIFICATION

Supervised Machine Learning: Regression and Classification, Stanford University and DeepLearning.AI (Coursera)

LANGUAGES

English: professional working proficiency; all four years of the BS were taught in English (Medium of Instruction certificate available). **Urdu:** native.

REFERENCES

Available on request.